η DECAY MODES

	Mode	Fraction (Γ_i / Γ)	Scale Factor/ Conf. Level	P(MeV/c)
Neutral modes				
Γ_1	neutral modes	(71.95 ± 0.29) %	S=1.3	
Γ_2	2γ	(39.36 ± 0.18) %	S=1.1	274
Γ_3	$3\pi^0$	(32.56 ± 0.21) %	S=1.2	179
Γ_4	$\pi^0 2\gamma$	$(2.55 \pm 0.22) \times 10^{-4}$		257
Γ_5	$2\pi^0 2\gamma$	$< 1.2 \times 10^{-3}$	CL=90%	238
Γ_6	4γ	$< 2.8 \times 10^{-4}$	CL=90%	274
Γ_7	invisible	$< 1.0 \times 10^{-4}$	CL=90%	
Charged modes				
Γ_8	charged modes	(28.05 ± 0.29) %	S=1.3	
Γ_9	$\pi^+\pi^-\pi^0$	(23.02 ± 0.25) %	S=1.2	174
Γ_{10}	$\pi^+\pi^-\gamma$	(4.28 ± 0.07) %	S=1.1	236
Γ_{11}	$e^+e^-\gamma$	$(7.00 \pm 0.22) \times 10^{-3}$	S=1.1	274
Γ_{12}	$\mu^+\mu^-\gamma$	$(3.1 \pm 0.4) \times 10^{-4}$		253
Γ_{13}	e^+e^-	$< 7 \times 10^{-7}$	CL=90%	274
Γ_{14}	$\mu^+\mu^-$	$(5.8 \pm 0.8) \times 10^{-6}$		253

$\omega(782)$ DECAY MODES

	Mode	Fraction (Γ_i / Γ)
Γ_1	$\pi^+\pi^-\pi^0$	(89.2 ± 0.7) %
Γ_2	$\pi^0\gamma$	(8.33 ± 0.25) %
Γ_3	$\pi^+\pi^-$	(1.53 ± 0.12) %
Γ_4	neutrals (excluding $\pi^0\gamma$)	$(7^{+7}_{-5}) \times 10^{-3}$
Γ_5	$\eta\gamma$	$(4.5 \pm 0.4) \times 10^{-4}$
Γ_6	$\pi^0 e^+ e^-$	$(7.7 \pm 0.6) \times 10^{-4}$
Γ_7	$\pi^0\mu^+\mu^-$	$(1.34 \pm 0.18) \times 10^{-4}$
Γ_8	$\eta e^+ e^-$	
Γ_9	e^+e^-	$(7.41 \pm 0.19) \times 10^{-5}$
Γ_{10}	$\pi^+\pi^-\pi^0\pi^0$	$< 2 \times 10^{-4}$
Γ_{11}	$\pi^+\pi^-\gamma$	$< 3.6 \times 10^{-3}$
Γ_{12}	$\pi^+\pi^-\pi^+\pi^-$	$< 1 \times 10^{-3}$
Γ_{13}	$\pi^0\pi^0\gamma$	$(6.7 \pm 1.1) \times 10^{-5}$

$\rho(770)^0$ decays

Γ_6	$\pi^+\pi^-$	~ 100 %
Γ_7	$\pi^+\pi^-\gamma$	$(9.9 \pm 1.6) \times 10^{-3}$
Γ_8	$\pi^0\gamma$	$(4.7 \pm 0.8) \times 10^{-4}$
Γ_9	$\eta\gamma$	$(3.00 \pm 0.21) \times 10^{-4}$
Γ_{10}	$\pi^0\pi^0\gamma$	$(4.5 \pm 0.8) \times 10^{-5}$
Γ_{11}	$\mu^+\mu^-$	[3] [4] $(4.55 \pm 0.28) \times 10^{-5}$
Γ_{12}	e^+e^-	[3] [4] $(4.72 \pm 0.05) \times 10^{-5}$
Γ_{13}	$\pi^+\pi^-\pi^0$	$(9 \pm 4) \times 10^{-5}$
Γ_{14}	$\pi^+\pi^-\pi^+\pi^-$	$(1.8 \pm 0.9) \times 10^{-5}$
Γ_{15}	$\pi^+\pi^-\pi^0\pi^0$	$(1.6 \pm 0.8) \times 10^{-5}$
Γ_{16}	$\pi^0 e^+ e^-$	$< 1.2 \times 10^{-5}$
Γ_{17}	$\eta e^+ e^-$	

$\phi(1020)$ DECAY MODES

	Mode	Fraction (Γ_i / Γ)	Sc
Γ_1	K^+K^-	(49.9 ± 0.5) %	
Γ_2	$K^0_L K^0_S$	(33.6 ± 0.4) %	
Γ_3	$\rho\pi + \pi^+\pi^-\pi^0$	(14.9 ± 0.4) %	
Γ_4	$\pi^+\pi^-\pi^0$		
Γ_5	$\eta\gamma$	(1.306 ± 0.024) %	
Γ_6	$\pi^0\gamma$	$(1.33 \pm 0.05) \times 10^{-3}$	
Γ_7	$\ell^+\ell^-$		
Γ_8	e^+e^-	$(2.964 \pm 0.033) \times 10^{-4}$	
Γ_9	$\mu^+\mu^-$	$(2.86 \pm 0.22) \times 10^{-4}$	
Γ_{10}	$\eta e^+ e^-$	$(1.08 \pm 0.04) \times 10^{-4}$	
Γ_{11}	$\pi^+\pi^-$	$(9.5 \pm 1.9) \times 10^{-5}$	
Γ_{12}	$\omega\pi^0$	$(4.7 \pm 0.5) \times 10^{-5}$	
Γ_{13}	$\omega\gamma$	<5 %	

→ Connect to zoom

STUDY NEUTRAL MESON DECAYS TO SYSTEM OF PIONS

- lab. to study
  - → STRONG INTERACTION
  - → SYMMETRIES (P, C, G, I)

RECAP OF CONSERVATION RULES

	P	C	G	I
STRONG	✓	✓	✓	✓
ELM	✓	✓	✗	✗
EWK	✗	✗	✗	✗

- P, C, G for $f\bar{f}$ systems (mesons)

$$P(f\bar{f}) = (-1)(1)(-1)^{L} = (-1)^{L+1}$$

$$C(f\bar{f}) = (-1)^{L+S} \leftarrow C = \mathcal{P} \times \mathcal{S}$$

$$G(f\bar{f}) = (-1)^{L+S+I} \leftarrow G = C R_2$$

· Lets build the $q\bar{q}$ multiplets.

$$J = L + S \begin{cases} J=0 \rightarrow L=0;\ S=0 \ \uparrow\downarrow \\ J=1 \rightarrow L=0;\ S=1 \ \uparrow\uparrow;\downarrow\downarrow \end{cases}$$

$L=0 \longrightarrow P=-1$ always

$S=0 \longrightarrow C=+1 \longrightarrow J^{PC}=0^{-+}$ pseud.

$S=1 \longrightarrow C=-1 \longrightarrow J^{PC}=1^{--}$ vect.

	J^{PC}	I	$G=(-1)^{J+I}$
η	0^{-+}	0	$+1$
ρ	1^{--}	1	$+1$
ω	1^{--}	0	-1
ϕ	1^{--}	0	-1

<u>NB</u> We focus only on neutral mesons b/c we want to study strong int. where $\mathbb{C}$ is conserved. $\mathbb{C}$ is defined only for

neutral particles.

- what do we measure experim.?

	2π	3π
η	×	30%
ρ	$\pi^-\pi^+$ 100% $\pi^0\pi^0$ 0%	×
ω	×	90%
ϕ	×	15%

- ρ only in 2π
- η, ω, ϕ only in 3π BUT VERY DIFFERENT BR!

- We study P, C, G for systems of n pions to understand experim. results.

RECALL → $\pi \Rightarrow 0^{-+}$; $G=-1$; $I=1$; $S=0$

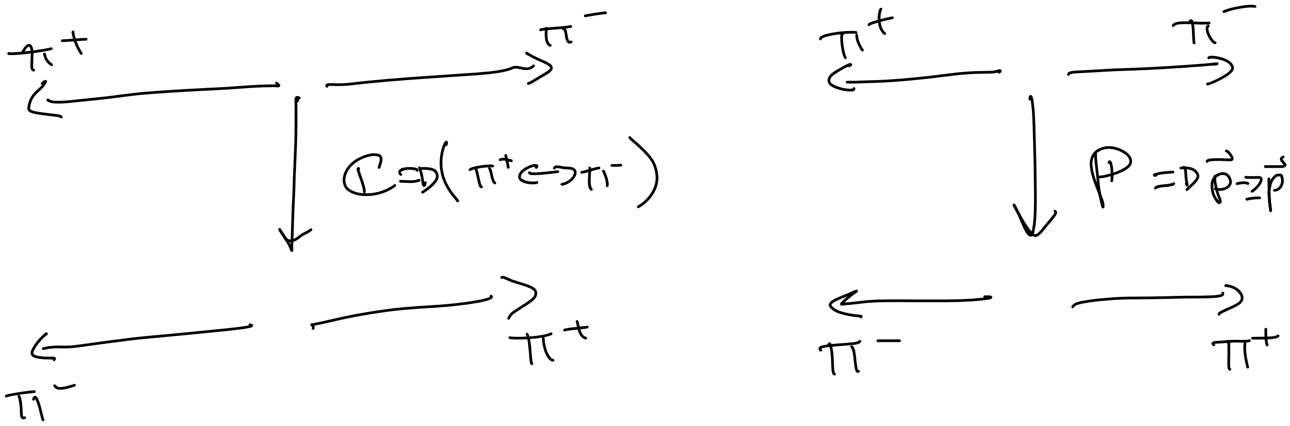
PARITY

$$P(n\pi) = (-1)^n (-1)^L$$

CHARGE CONJ.

$C(\pi^0) = +1 \qquad C(\pi^+) = \pi^- \qquad C(\pi^-) = \pi^+$

$C(\pi^+\pi^-) = ?$



$$C(\pi^+\pi^-) = P(\pi^+\pi^-) = (-1)^\ell$$

If $n > 2$ C cannot be evaluated w/o considering I.

G-PARITY

$$G = C(-1)^I$$

$$G(\pi^0\pi^0) = G(\pi^+\pi^-) = (-1)^\ell \cdot (-1)^I$$

$\hookrightarrow |1,+1\rangle |1,-1\rangle =$

$= \alpha|0,0\rangle + \beta|1,0\rangle + \gamma|2,0\rangle$

$\alpha, \beta, \gamma \neq 0$

$\hookrightarrow |1,0\rangle|1,0\rangle = \alpha'|0,0\rangle + \gamma'|2,0\rangle$

$\alpha', \gamma' \neq 0$

<u>NB</u> $\pi^0\pi^0$ and $\pi^+\pi^-$ have different I_{TOT} when combined.

$\hookrightarrow$ When studying STRONG INT. where I has to be conserved initial and final state have to have SAME I!

if $n > 2$ generalize:

$$G(n\pi) = (-1)^n$$

Now we study η, ρ, ω, ϕ decays:

$\eta \leftrightarrow 0^{-+}$; $G = +1$; $J = 0 = L + S$

I need to conserve J:

$S = 0$ (system of mesons)

If $J = 0 \to L = 0$

- $\eta \to 2\pi$

$$G(2\pi) = (-1)^2 = +1 \checkmark$$

$$P(2\pi) = (-1)^2 = +1 \times$$

Forbidden b/c P violated!

- $\eta \to 3\pi$

$$G(3\pi) = (-1)^3 = -1 \times$$

$$P(3\pi) = (-1)^3(-1)^0 = -1 \checkmark$$

$C(3\pi)$ from:

$$\underset{-1}{G} = C\,(-1)^{I}$$

if $I = 1$

$\Rightarrow C = 1 \checkmark$

Forbidden strong b/c G violated
but since C, P ✓ it happens
ELM! BR ~ 35%

$\rho \longleftrightarrow 1^{--}$; $G = +1$; $J = +1$

$\downarrow$

$L = +1$

$\rho \to 2\pi$

$G(2\pi) = +1$ ✓

$P(2\pi) = (-1)^2(-1)^L = -1$ ✓

$C(2\pi) = -1$ ✓

$Q = m_\rho - 2m_\pi \sim 600\,\text{MeV} > 0$ ✓

it happens STRONG BUT:

$BR(\pi^0\pi^0) = 0\ \%$

$BR(\pi^+\pi^-) = 100\%$ → why?

$\Psi(\pi^0\pi^0) = \Psi_S \cdot \Psi_s \cdot \Psi_I$

has to be symm. $1 \leftrightarrow 2$ b/c. B-E stat

Ψ_S $\to$ has to be symm.

Ψ_I $\to$ symm. $\to |1,0\rangle|1,0\rangle = \alpha|0,0\rangle + \beta|2,0\rangle$ symm

...but $L=1$ $\to$ Ψ_S gets $(-1)^L$ hence antisymm.

$\Rightarrow \pi^0\pi^0$ decay is suppressed.

· $\rho \to 3\pi$

$G(3\pi) = -1$ ✗

$P(3\pi) = +1$ ✗

Forbidden b/c G VIOLATED!

Differently from $\omega \to 3\pi$, also P is violated. $\to$ NO ELM DECAY.

$\omega \longleftrightarrow 1^{--}$; $G = -1$; $S=1 \rightarrow C=1$

· $\omega \rightarrow 2\pi$

$$G = (-1)^2 = +1 \times$$

Forbidden, G violated!

· $\omega \rightarrow 3\pi$

$$G(3\pi) = -1 \checkmark$$

STRONG, BR ~ 89%

Same happens for ϕ b/c also ϕ is 1^{--} ; $G = -1$ but why VERY DIFFERENT BR w.r.t. ω?

NB ϕ is a composition of s quarks while ω of u,d.

$\rightarrow$ final state w/ pions has u, d.

$\rightarrow \phi' \rightarrow 3\pi$ suppressed b/c.
"disconnected diagram"
or OZI rule

you will see it
soon :)